

Volumes from Known Cross Sections and Cylindrical Shells

Directions

- Every length in these problems is a real measurement. The units are stated in each problem; every volume is therefore a cubic unit, and the answer line names which one.
- Write the integral before you evaluate it: state the cross-sectional area $A(x)$, or the shell radius and height, and the limits.
- Give an exact answer (a fraction, a multiple of π , or a radical) and then a decimal to two places.

1. The base of a solid is the region in the first quadrant bounded by the line $y = x$, the x -axis, and the line $x = 4$, where x and y are measured in metres. Every cross section perpendicular to the x -axis is a square with one side in the base. Find the volume of the solid.

Answer: _____ m^3

2. The base of a solid is the region bounded by $y = \sqrt{x}$, the x -axis, and the line $x = 9$, where x and y are in centimetres. Every cross section perpendicular to the x -axis is a semicircle whose diameter runs from the x -axis to the curve. Find the volume of the solid.

Answer: _____ cm^3

3. The region bounded by $y = x^2$, the x -axis, and the line $x = 3$, with x and y in feet, is revolved about the y -axis. Use cylindrical shells to find the volume of the resulting solid.

Answer: _____ ft^3

4. The base of a solid is the triangular region bounded by the line $y = 4 - x$ and the two coordinate axes, where x and y are in centimetres. Every cross section perpendicular to the x -axis is an equilateral triangle with one side in the base. (Recall that an equilateral triangle of side s has area $\frac{\sqrt{3}}{4}s^2$.) Find the volume of the solid.

Answer: _____ cm^3

5. The region bounded by $y = 4 - x^2$ and the x -axis, for $x \geq 0$, is revolved about the y -axis. All lengths are in metres. Use cylindrical shells to find the volume of the resulting solid.

Answer: _____ m^3

6. The region bounded by $y = \sqrt{x}$, the x -axis, and the line $x = 9$ (lengths in centimetres) is revolved about the y -axis.

- (a) Use cylindrical shells to find the volume of the solid.
- (b) Explain, in one or two sentences, why the shell setup uses the curve exactly as written while a washer setup would first have to rewrite it.

Answer: _____ cm^3

7. The base of a solid is the region enclosed by the curves $y = x$ and $y = x^2$, with x and y in metres. Every cross section perpendicular to the x -axis is a square.

(a) Find the two x -values where the curves meet.

Answer: _____

(b) Find the volume of the solid.

Answer: _____ m^3

8. The region bounded by $y = x^2$, the x -axis, and the line $x = 2$ (lengths in centimetres) is revolved about the vertical line $x = 4$.

(a) Use cylindrical shells to find the volume of the resulting solid.

(b) Explain why the shell radius is $4 - x$ and not x , and check your expression at both ends of the interval.

Answer: _____ cm^3